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What to expect from opening up 'black boxes'? Comparing perceptions of justice between human and automated agents

Nadine Schlicker [a,c,d,*], Markus Langer [a], Sonja K. Ötting [b], Kevin Baum [a], Cornelius J. König [a], Dieter Wallach [c]

[a] Fachrichtung Psychologie, Universität des Saarlandes, Saarbrücken, Germany
[b] Fachrichtung Psychologie, Universität Bielefeld, Bielefeld, Germany
[c] Ergosign GmbH, Saarbrücken, Germany
[d] Institut für Künstliche Intelligenz in der Medizin, Philipps-Universität Marburg, Marburg, Germany

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ABSTRACT

Advances in artificial intelligence contribute to increasing automation of decisions. In a healthcare-scheduling context, this study compares effects of decision agents and explanations for decisions on decision-recipients' perceptions of justice. In a 2 (decision agent: automated vs. human) × 3 (explanation: no explanation vs. equality-explanation vs. equity-explanation) between-subjects online study, 209 healthcare professionals were asked to put themselves in a situation where their vacation request was denied by either a human or an automated agent. Participants either received no explanation or an explanation based on equality or equity norms. Perceptions of interpersonal justice were stronger for the human agent. Additionally, participants perceived human agents as offering more voice and automated agents as being more consistent in decision-making. When given no explanation, perceptions of informational justice were impaired only for the human decision agent. In the study's second part, participants took the perspective of a decision-maker and were given the choice to delegate decision-making to an automated system. Participants who delegated an unpleasant decision to the system frequently externalized responsibility and showed different response patterns when confronted by a decision-recipient who asked for a rationale for the decision.

1. Introduction

Systems that use methods from the broad area of artificial intelligence (AI)[1] increasingly spread throughout various application areas. Such systems now automate decision-making in tasks as diverse as medical diagnosis, jurisprudence, and management (Adadi & Berrada, 2018; Ghislieri et al., 2018; Grgić-Hlača et al., 2019; Longoni et al., 2019). Automated decision agents are able to outperform human decision-making capabilities, especially in rule-based and repetitive tasks that require high levels of attention and cognitive capacity (Chowdhury & Sadek, 2012; Haenssle et al., 2018). Commonly acknowledged possible advantages of automated decision-making are higher processing speed (Haenssle et al., 2018), less bias (Kleinberg et al., 2018), and generally more justice in the distribution of outcomes (M. K. Lee, 2018; Ötting & Maier, 2018). Such advantages regarding actual justice but also regarding justice perceptions would be important to everyday decisions within organizations, where justice perceptions positively affect employees' job motivation, satisfaction and intentions to remain loyal to their employer (Cohen-Charash & Spector, 2001). These perceptions are grounded in the evaluation of justice regarding the distribution of outcomes, decision processes, and in an evaluation of interpersonal treatment and the transparency in decision-making (Dzindolet et al., 2003; M. K. Lee, 2018; Uhde et al., 2020) – factors that reflect the four facets of organizational justice: distributive, procedural, interpersonal, and informational justice (Colquitt, 2001).

Although there is hope that automated systems improve justice of organizational decision processes and outcomes in comparison to decisions made by human decision agents, previous findings are far from

* Corresponding author. Institut für Künstliche Intelligenz in der Medizin, Universitätsklinikum Gießen und Marburg GmbH, Baldingerstraße, 35043, Marburg, Germany.
E-mail address: nadine.schlicker@uni-marburg.de (N. Schlicker).

[1] We use the term „artificial intelligence (AI)" to refer to both human-coded algorithms as well as algorithms based on machine learning and deep artificial neural networks.

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conclusive and it might be necessary to take a more nuanced view on the facets of justice (M. K. Lee & Baykal, 2017; Waytz & Norton, 2014). Specifically, there is research suggesting that automated systems might contribute to more consistent decision processes and outcomes (Langer et al., 2019), whereas other findings imply that automated systems reduce human involvement in decision processes which can detrimentally affect perceptions of interpersonal justice (Acikgoz et al., 2020), and yet other research found no differences regarding justice perceptions between human and automated decision agents (Ötting & Maier, 2018). These results indicate that automated decision-making might improve certain facets of justice but detrimentally affect others. However, there is only scarce research that has investigated justice perceptions in regard to automated decisions and the few existing studies focused on general perceptions of fairness as an overall outcome of justice (Langer et al., 2018; M. K. Lee, 2018; Newman et al., 2020) or on single facets of justice (Ötting & Maier, 2018). Consequently, there is a lack of studies investigating for which of the four facets of justice automated decisions might be beneficial and for which they might be obstructive.

Irrespectively of the decision agent, perceptions of justice also depend on information and explanation in regard to decision processes and outcomes (Dodge et al., 2019; Greenberg & Colquitt, 2013). Specifically, providing explanations of decision outcomes should directly affect whether people believe that they receive honest and open treatment (i.e., contributing to perceptions of informational justice). Furthermore, explanations can indirectly affect perceptions of justice because they enable understanding of processes and respective decision outcomes (Greenberg & Colquitt, 2013; Skarlicki et al., 2008). As this study compares the effects of explanations in human-human and human-automation interaction we use a definition of explanations that can be applied to both types of interactions. We define explanation here as the provision of "information about [an event's] causal history. In an act of explaining, […] [an agent] who is in possession of some information about the causal history of some event […] tries to convey it to someone else" (Lewis, 1986, p. 217).

Whereas humans can easily provide information and explanations regarding decision processes, the same does not necessarily apply to automated systems (Burrell, 2016). Specifically, automated systems increasingly rely on AI approaches that make it hard (e.g., due to large numbers of predictors or opaque algorithms) or even impossible to understand the systems' internal processes (Adadi & Berrada, 2018; Rahwan et al., 2019). As a result, people responsible for decision-making and people affected by automated decisions might have no insight into why a system produced a specific output and made a respective decision. This issue amplifies with the spread of automated systems into new application areas as people operating automated systems or confronted with automated decisions have little understanding of how automated systems arrive at their outputs (Burrell, 2016) which can impair their perceptions of justice (Dodge et al., 2019; M. K. Lee et al., 2019). Moreover, even if automated systems could provide adequate explanations regarding their outputs, it is not clear whether findings regarding the effects of information and explanation that predominantly stem from the context of human decision-making (Shaw et al., 2003; Truxillo et al., 2009) translate to the context of automated decision-making (Langer et al., 2018; Newman et al., 2020).

This study examines the effects of decision agents and explanations in a scheduling context among healthcare providers (e.g., nurses, physicians). For a holistic view on justice in decision-making, we examine how the decision agent (human vs. automated) affects people's perceptions of the four justice facets (distributive, procedural, informational and interpersonal justice, Colquitt, 2001) with regard to a scheduling decision. We further investigate the effects of different explanations (Deutsch, 1975) and the absence of any explanation accompanying the decision with respect to human versus automated decision agents. In the exploratory second part of this study, we consider the perspective of the responsible decision-makers shedding light onto people's choice to delegate unpleasant decisions to automated systems.

2. Theoretical background

2.1. Automation in scheduling

Making decisions in sensitive contexts affecting people's everyday life has traditionally been reserved to human decision agents. This is due to the fact that decision-making in such domains can be complex (Mittelstadt & Floridi, 2016). Decision-makers need to consider individual interests and needs, and try to make the process and outcomes as just as possible. One example are work scheduling decisions constituting classical decision-making situations usually handled by human decision-makers (Constantino et al., 2015; Uhde et al., 2020). Work-scheduling considers regulation (e.g., minimum breaks required between shifts), workforce demands, employees' contracts, as well as employees' needs. It is also important that scheduling processes are just and that shifts are distributed in a just way (Sharma & Dhar, 2016).

However, recent advances in technology contribute to the growing possibility to automate decisions traditionally reserved for human decision-makers. Specifically, easier availability of large data sets, cheaper storage capacities, higher computing power, and progress in mathematical methods underlying algorithm-based systems make automated decision-making more likely (Schmidt et al., 2019). Automation increases even in sensitive and complex decision contexts affecting individual employees, such as scheduling (Grgić-Hlača et al., 2019; Wesche & Sonderegger, 2019). Furthermore, also ethical considerations can now be taken into account within automated decision-making. For instance, automated shift scheduling systems can support the creation of schedules adhering to standards that follow various mathematical definitions of fairness (Constantino et al., 2015; Lin et al., 2015). Fairness, however, cannot be reduced to a mathematical concept, but also includes components such as the perceived justice of transparency and respect (Colquitt et al., 2001).

2.2. Justice

Regardless of the decision agent, decision-making involves questions concerning justice, like employee demands regarding individual considerations, just processes, transparency and just distributions of outcomes (Colquitt et al., 2001; Ötting & Maier, 2018). Justice is defined as the "perceived adherence to rules that reflect appropriateness in decision contexts" (Colquitt & Rodell, 2015, p. 188). These rules apply to four facets: interpersonal, procedural, distributive, and informational justice. Interpersonal justice refers to treating individuals with respect and dignity (Colquitt et al., 2001). Procedural justice describes the justice of processes leading to a decision outcome. It consists of perceptions that processes are unbiased, correctable, accurate, and ethical (Colquitt et al., 2001). Furthermore, perceptions of procedural justice are stronger when individuals have voice in a decision process (Folger, 1977; Lind et al., 1990). Distributive justice concerns the perceived justice of the distribution of outcomes and is given, when the allocation of an outcome is perceived to be appropriate for a particular situation (Colquitt, 2001). Finally, informational justice relates to the justification and the adequacy of explanations for procedures in terms of timeliness, specificity, and faithfulness (Colquitt, 2001).

2.2.1. Comparing perceptions of justice between human and automated agents

People's justice perceptions regarding human and automated decision agents might not be unanimously positive or negative. For instance, in an experimental vignette study (in a task allocation scenario), Ötting and Maier (2018) manipulated dimensions of procedural justice (e.g., a decision process was described as inconsistent) and the type of decision agent (human, humanoid robot, or computer), and investigated effects on participants perceptions (e.g., imagined job satisfaction in the

situation described in the vignette). Although treated as a manipulation check in their study, they also investigated whether there was a difference for the decision agent for the procedural justice scale. For both of their studies they did not find an effect between human and automated agents for procedural justice. Other previous research manipulated the decision agent in school admission scenarios and found automated agents to be perceived as more just than human agents (Marcinkowski et al., 2020), but the respective study only investigated an overall assessment of justice instead of treating justice as a multifaceted construct. Yet other research suggests negative justice perceptions with respect to automated agents in job interview scenarios (Acikgoz et al., 2020; Langer et al., 2019b) where most of the negative effects were found for variables associated with interpersonal justice. These mixed findings led to our proposition that the specific strengths and weaknesses of the respective decision agent with respect to interpersonal, procedural, distributive, and informational justice contribute to positive and negative justice perceptions in different decision contexts.

The strength of human decision agents lies in building interpersonal relationships with decision-recipients. Humans are able to consider individual background information and show interpersonal warmth in decision processes (M. K. Lee & Baykal, 2017). In cases where decision-recipients experience unfavorable circumstances, humans are able to provide individualized apologies. In comparison, automated systems are perceived as less able to show empathy or adapt to people's emotions (M. K. Lee, 2018). They also perform worse in tasks that require emotional skills (Waytz & Norton, 2014). All of this could contribute to the fact that people perceive less interpersonal warmth and individual consideration when confronted with automated decisions (Langer et al., 2019).

The strength of automated decision agents may lie in their consistency, higher processing capacity and efficiency (M. K. Lee, 2018) with implications for procedural, distributive, and informational justice. Perceptions of procedural justice might be positively affected by the higher consistency of processes of automated agents. Specifically, current automated systems will use the same input information and execute the same processes for every decision and are not influenced by factors such as fatigue or a lack of motivation (Bigman et al., 2020; Bigman & Gray, 2018). Especially, in the case of cognitively straining repeated decision-making demands, human information processing becomes more superficial and hence may result in (unintended) unfair or suboptimal outcomes (Tversky & Kahneman, 1974). In contrast, automated systems have higher processing capacity and (at least in the case of non-learning systems) the same inputs will result in the same outputs. Thus, they provide higher consistency of procedures, which might let people believe that automated systems are inherently more objective and consequently can contribute to higher perceived procedural justice of decision processes (Langer et al., 2019; F. A. Miller, Katz, & Gans, 2018).

Distributive justice could benefit from automated systems' higher processing capacity and efficiency. Specifically, automated systems' ability to perform complex analyses on large datasets, compare individual contributions and efforts, or to divide resources equally among people might be seen as superior (Dzindolet et al., 2003; M. K. Lee et al., 2019; Mouloua, 2018; Skitka et al., 2000). Consequently, people might assume that systems will optimally distribute outcomes (M. K. Lee et al., 2019) resulting in higher distributive justice perceptions.

In the case of informational justice, people's perceptions of automated systems could also benefit from higher consistency. More precisely, automated systems do not have motives, interests or emotions that could interfere with this consistency (M. K. Lee, 2018). An explanation by an automated agent could thus be perceived as thorough since people believe them to be based on mathematical procedures and to have consistently considered available data (Pessach & Shmueli, 2020). People might further believe that systems are inherently "honest" since they have no motivation to manipulate or lie (Bigman et al., 2019) whereas humans may be motivated to distort an explanation based on their goals and motivations (e.g., to hide their real aims or to maintain interpersonal relationships) (van der Werff et al., 2019). Consequently, informational justice could be stronger in the automated agent condition.

In sum, the aforementioned arguments for potential differences in justice perceptions between automated and human decision agents condense in the following hypotheses:

H1a. Participants perceive stronger interpersonal justice in the human compared to the automated agent condition.

H1b. Participants perceive stronger procedural justice in the automated compared to the human agent condition.

H1c. Participants perceive stronger distributive justice in the automated compared to the human agent condition.

H1d. Participants perceive stronger informational justice in the automated compared to the human agent condition.

2.2.2. The influence of explanations on justice

Explanations are strongly tied to justice perceptions. For instance, prior research found positive effects of managers'' explanations on employees' perceptions of justice (Greenberg & Colquitt, 2013; Skarlicki et al., 2008). Similarly, explanations positively relate to fairness perceptions in human-computer interaction (Dodge et al., 2019; M. K. Lee et al., 2019).

Automated and human explanations come with different faces. One of the main differences is adaptability. Whereas humans try to tailor explanations to decision-recipients and adapt explanations within conversations, automated explanations are (to date) mostly fixed to the specific programming or underlying algorithm (T. Miller, 2019). Although, there are attempts to make automated agents more interactive (Cawsey, 1991; Guzman & Lewis, 2020), with the current state of technology, human-like dynamic explanations are not yet possible.

Explanations and information should directly affect perceptions of informational justice (Colquitt, 2001; Greenberg & Cropanzano, 1993; Skarlicki et al., 2008). This is due to the fact that informational justice perceptions concern the adequacy of explanations and information (Colquitt, 2001). Withholding or even denying explanations and information about procedures might be seen as inadequate and lacking transparency which should decrease perceptions of informational justice (Colquitt, 2001; Dodge et al., 2019).

The absence of an explanation might indirectly affect procedural, distributive and interpersonal justice perceptions (Cropanzano et al., 2015; Greenberg & Colquitt, 2013). More precisely, referring back to the definition of justice as "the perceived adherence to rules that reflect appropriateness in decision contexts" (Colquitt & Rodell, 2015, p. 188), it is plausible that it is harder to evaluate justice in the absence of explanations. Without an explanation accompanying a decision, the adherence to rules cannot be evaluated against a person's own standards. For instance, procedural justice refers to the perception that processes are consistent, unbiased, and ethical (Colquitt et al., 2001). Hence, procedural justice might decrease in the absence of an explanation, because decision-recipients cannot estimate these criteria. Similarly, the absence of an explanation might decrease distributive justice because the adequacy of the distribution of outcomes cannot be assessed. For instance, decision-recipients might not know whether chances were equal for all involved parties. Interpersonal justice refers to respectful and proper treatment of individuals (Colquitt et al., 2001). If an agent denies an explanation, recipients may perceive this as disrespectful and impolite thus reducing interpersonal justice (Bobocel & Zdaniuk, 2005).

Not only the presence or absence of an explanation can affect justice perceptions, but also the content of available explanations (Dhaliwal & Benbasat, 1996). For instance, in scheduling explaining that earlier shifts are preferably assigned to parents might raise different perceptions of justice in employees with and without family obligations (Uhde et al., 2020; Young, 1999). Explanations of the allocation of decision

outcomes can be especially relevant for perceived distributive justice. Deutsch (1975) distinguished three allocation norms of distributive justice: equity, equality, and need. Equity describes outcomes that are perceived as just when they are allocated corresponding to individual contributions. In scheduling this means that those who contribute more (e.g., work overtime), deserve more or relatively better outcomes. Equality describes the equal distribution of outcomes regardless of individual contributions and preconditions. In scheduling this means that every person has the same chance to receive their preferences. Need describes situations where outcomes are distributed according to individual situations and preconditions (e.g., considering family obligations). For example, in scheduling this means that parents might work early shifts, when childcare is guaranteed. If a distribution of outcomes fits the expected allocation norm of the decision-recipient, distributions will more likely be perceived as just (Colquitt, 2001; Deutsch, 1975). This means when organizations aim for just distributions they need to analyze the preferred allocation norms of their employees. Which allocation norm is applied depends on group goals (e.g., social relations vs. productivity), values (common vs. individual identity), and the specific context (e.g., work vs. family) (Ahmad, 2004; Deutsch, 1975).

Scheduling offers many different situations that require decisions, for example, how to allocate shifts in a schedule or how to permit vacation in the case of conflicting demands. In a study on healthcare workers' preferences with regard to the distribution of shifts in scheduling, Uhde et al. (2020) found that for the initial allocation of shifts, the equality norm was preferred. In the case of overlapping wishes or demands, healthcare professionals preferred to resolve these in direct communication on the basis of the need allocation norm. However, comparing needs might not always be self-evident and due to commonly conflicting needs mutual agreement is not guaranteed. For example, each person might claim that they have the need to spend Christmas with their family. One might claim that their need is higher because they have younger children and the other might claim that their children live abroad and only come home for Christmas. Such needs cannot be weighed easily against each other and lead to cases where a third party (e.g., a human supervisor or an automated system) may need to resolve the conflict.

Previous research has attempted to outline contexts and social groups where different allocation norms are preferred. For example, Uhde et al. (2020) found that in scheduling contexts the equity norm was either not mentioned or explicitly rejected by their participants. They assume that equity is perceived as less just because individual contributions are not easily and objectively comparable (e.g., due to different worktime models or individual preconditions). M. K. Lee et al. (2019) found that in a resource allocation task, equality was preferred by the majority of participants. Moreover, Lerner (1975, 1977) argues that the equality norm is more likely to be applied when parties expect common future interactions and healthcare professionals usually work in teams which makes future interactions likely.

Overall, in relation to explanations and justice perceptions we thus propose the following hypotheses:

H2. Participants perceive stronger justice when an explanation is present than when it is absent.

H3. Participants perceive stronger distributive justice when they receive an explanation that aligns with the allocation norm of equality rather than equity.[2]

[2] To support the call for open science (Open Science Collaboration, 2015), we preregistered hypotheses, research questions, operationalization and statistical analyses under http://aspredicted.org/blind.php?x=hn7z33. Hypothesis 5 in the preregistration corresponds to Hypothesis 3 in this paper included all dimensions of justice. However, while considering the literature in more depth, we realized that allocations norms mainly refer to the distribution of goods which justified only this specific hypothesis.

2.3. Possible interaction effects between the decision agent and explanations

We have outlined, that explanations and decision-agents are closely related to justice perceptions. Additionally, effects of decision agents and explanations are strongly intertwined. Recipients use decision agents' explanations to evaluate agent characteristics and intentions (Bobocel & Zdaniuk, 2005). Similarly, characteristics of the decision agent might also translate to the evaluation of the explanations provided by the respective decision agent (Bobocel & Zdaniuk, 2005; Shapiro et al., 1994).

Insights on the effects of explanations on justice perceptions predominantly come from investigating human decision agents' behavior and it is not yet clear whether those effects also apply to automated agents. For instance, attributions regarding the absence of an explanation may differ between human and automated decision agents (Burton et al., 2020; Elsbach & Stigliani, 2019). For an automated agent, the absence of an explanation might be attributed to technical issues or the complexity of data whereas the same situation with a human agent may be attributed to the agent's motivation to hide their true intentions (Kim & Hinds, 2006; van der Werff et al., 2019). In contrast, automated decisions without explanation could be perceived as hiding responsibilities behind a system or as an attempt to undermine potential complaints, which might more strongly decrease perceptions of justice compared to the absence of an explanation in human decision-making. However, as there is only scarce prior research that could provide rationales for specific hypotheses, we propose the following research question:

Research Question (RQ) 1: Are there differences regarding justice perceptions between the automated and the human agent depending on the presence and/or the type of explanation?

2.4. The perspective of the decision-maker

So far, our focus has been on the effects of decision agents and explanations on decision-recipients. However, it is necessary to consider all stakeholders involved in contexts where automated systems are on the verge of transforming existing decision processes (Floridi et al., 2018; Rahwan et al., 2019). Especially in healthcare scheduling, decision-makers often play a double role: They are responsible for the schedule and also take part in the care of patients. This makes the decision-maker a decision-recipient (e.g., team member) and a decision-maker (e.g., supervisor) at the same time, which can lead to role conflicts. In cases where several employees have the same needs, decision-makers cannot fulfill all employees' interests. In such cases, it is the decision-makers' duty to resolve impasses. When conflicts arise, the decision to reject or permit employees' wishes may be emotionally straining and unpleasant, because each outcome is connected to the resentment of disadvantaged colleagues (Kubek et al., 2019; Uhde et al., 2020). This may lead to anger and accusations against colleagues or decision-makers. When such sensitive decisions are made, it is likely that employees who end up with an unfavorable outcome might ask for an explanation (Brockner et al., 1997; Krehbiel & Cropanzano, 2000).

In cases where humans give up their decision-making autonomy by delegating decisions to automated systems, those systems could relieve humans from the burden of unpleasant decision-making (Floridi et al., 2018). As a result of reduced decision-making autonomy, human agents could (partially) externalize responsibility to a system, thus feeling less responsible for a decision outcome (Bigman et al., 2019; Matthias, 2004; Morgeson et al., 2012). Delegating responsibility to an automated agent, might also provide a possible justification when decision-recipients are dissatisfied (Hackman & Oldham, 1976; Morgeson et al., 2012). Since there are only initial assumptions regarding possible differences in delegating unpleasant decisions to automated systems and consequential effects on decision-makers (Bigman et al., 2019), the second part of the study exploratory investigates those assumptions. By asking participants to consider the perspective of the decision-maker, we pose the

following explorative research questions:

RQ 2a: Do individuals who are put in the position of a decision-maker prefer to have the authority to make decisions themselves or do they prefer to delegate decisions to a system?

RQ 2b: Do preferences for retaining or delegating decisions depend on the decision agent that previously denied the participant's request for vacation?

RQ 3a: How do participants respond to the employee who asks why she was not granted vacation leave?

RQ 3b: Do these responses differ when participants retained vs. delegated the decision-making?[3]

3. Method

3.1. Sample

We calculated the required sample size using G-Power (Faul et al., 2007). We assumed a low to moderate effect (partial $\eta^2 = 0.04$) (Richardson, 2011). Accordingly, $N = 191$ participants were necessary to detect main effects and interactions in a 2 × 3 between-subjects ANOVA with a power of .80. As the effect size estimation was rather liberal, we decided to include more than 191 participants if possible. Data was collected in Germany via snowball sampling in social media groups for health care workers, on bulletin boards in healthcare institutions, and via direct contact with hospitals, retirement homes or employees in the healthcare sector. The recruited sample consisted of $N = 217$ participants. Since we collaborated with a company involved in the development of automated scheduling systems, participants were excluded if they were employed at an institution where the prototype of the scheduling system was developed ($n = 1$). Further, participants were excluded if they indicated that their data should not be used for analysis ($n = 4$), if they were not employed within the healthcare sector ($n = 2$), and if they took longer than 1 h to read the story and answer the first set of questions ($n = 2$) indicating that they temporarily walked away from the study. The final sample consisted of $N = 209$ participants (83.3% female) (Table 2 shows the number of participants in the experimental conditions). Of the participants, 50.2% were employed in hospitals, 23.9% in geriatric institutions, 12.0% in outpatient care, 2.9% in homes for the disabled and 11.0% named more than one institution or a specific healthcare institution. Participants mean age was $M = 35.25$ years ($SD = 11.86$); mean tenure was $M = 11.57$ years ($SD = 10.96$), 45.9% of the participants declared that they had already been in charge of scheduling during their employment and 84.7% of the participants were engaged in shift work.

3.2. Procedure

We conducted the study online and used a randomized 2 (decision agent: automated vs. human) × 3 (explanation: equality-explanation vs. equity-explanation vs. no explanation) between-subjects design. The study consisted of two parts (see Fig. 1 for an overview). Participants should imagine being the person in the story. In part one, participants were in the position of a decision-recipient and it covered all hypotheses (H1a to H3) and RQ 1. In part two, participants were in the position of a decision-maker and it covered the explorative research questions RQ 2 and RQ 3. In the beginning of the survey, participants provided their informed consent and responded to demographic questions. Participants then proceeded to the introduction page of the story. Throughout the story, the experimental manipulations were realized. The story was depicted in comic style[4] (see Figs. 2 and 3) to improve psychological fidelity (Aguinis & Bradley, 2014; Diefenbach & Hassenzahl, 2017; Taylor, 2005).

In the beginning of the story, participants were instructed to imagine being a nurse in a hospital and received information helping them to immerse in the story (Scene 1, Fig. 1; Fig. 2). Then, participants were introduced to the reason for their vacation request (i.e., being invited to a wedding; Scene 2, Fig. 1). In Scene 3, we varied the first independent variable (decision agent: human vs. automated). Specifically, participants were presented with pictures and text describing that they applied for vacation with either a human or an automated system. Scenes 4 and 5 then highlighted that the employees have the opportunity to try to find a solution themselves in line with their respective needs. Specifically, in Scene 4, participants were informed that the application for vacation could not be granted because another colleague has applied for vacation at the same time period. Participants were advised to contact their colleague to find a joint solution. In the human agent condition this message was delivered by the supervisor. In the automated agent condition this message was displayed on a technical device. In Scene 5, participants were told that their colleague is not personally available and therefore they had to text them on their smartphone. This was done to ensure that there were no differences in how the communication with their colleague occurred in the human and the automated agent condition. During the conversation with their colleague it becomes apparent that they cannot reach mutual agreement. In Scene 6, participants read that they either told their supervisor (human-agent condition) or entered in the system (automated-agent condition) that they and their colleague could not reach mutual agreement. The respective decision agent responded that they will decide and inform them after their colleague has confirmed that no agreement was reached. This information was provided to ensure that participants assume that the agents also consider their colleague's perspective. Otherwise, participants might have thought that they could have provided any information without both parties being consulted.

In Scene 7, participants were informed that their vacation request has been denied. This information was either delivered by the supervisor (human agent) or displayed on the technical device (automated agent). In this scene, the second independent variable (explanation) was varied (see Fig. 3). The reason for denying the vacation was either explained based on the allocation norm of equality reflected by a decision based on randomization ("Unfortunately, your vacation request has been denied. Your colleague has been chosen at random to receive their requested vacation") or based on the allocation norm equity reflected by the number of hours of overtime of each employee ("Unfortunately, your vacation request has been denied. Your colleague has been chosen based on past data. They worked more overtime than you"). In the no explanation condition, the reason for the decision was not explained ("Unfortunately, your vacation request has been denied. Your colleague has been chosen to receive their vacation"). Following this part of the study, participants responded to items assessing justice perceptions (Colquitt, 2001) and perceived understandability of the decision agent (Madsen & Gregor, 2000).

For the second part of the study, participants were again asked to imagine they would be the nurse from part one of the story. Participants were informed that after some time, they had been promoted to the position of a decision-maker responsible for scheduling in the same hospital (Scene 8). Participants in the human decision-agent condition were then introduced to the automated system for scheduling (i.e., received the information about the automated system that the other group received in part one of the story) and participants in the system decision-agent condition were told about the old scheduling method

[3] In the preregistration, we mention hypotheses regarding trust in the decision agent. However, we realized that the rationale for examining trust within the current design was weak. We thus decided to focus on justice. Data and results for the trust measures are available upon request.

[4] All pictures and the corresponding descriptions for each group can be found in Appendix A of the supplementary material.

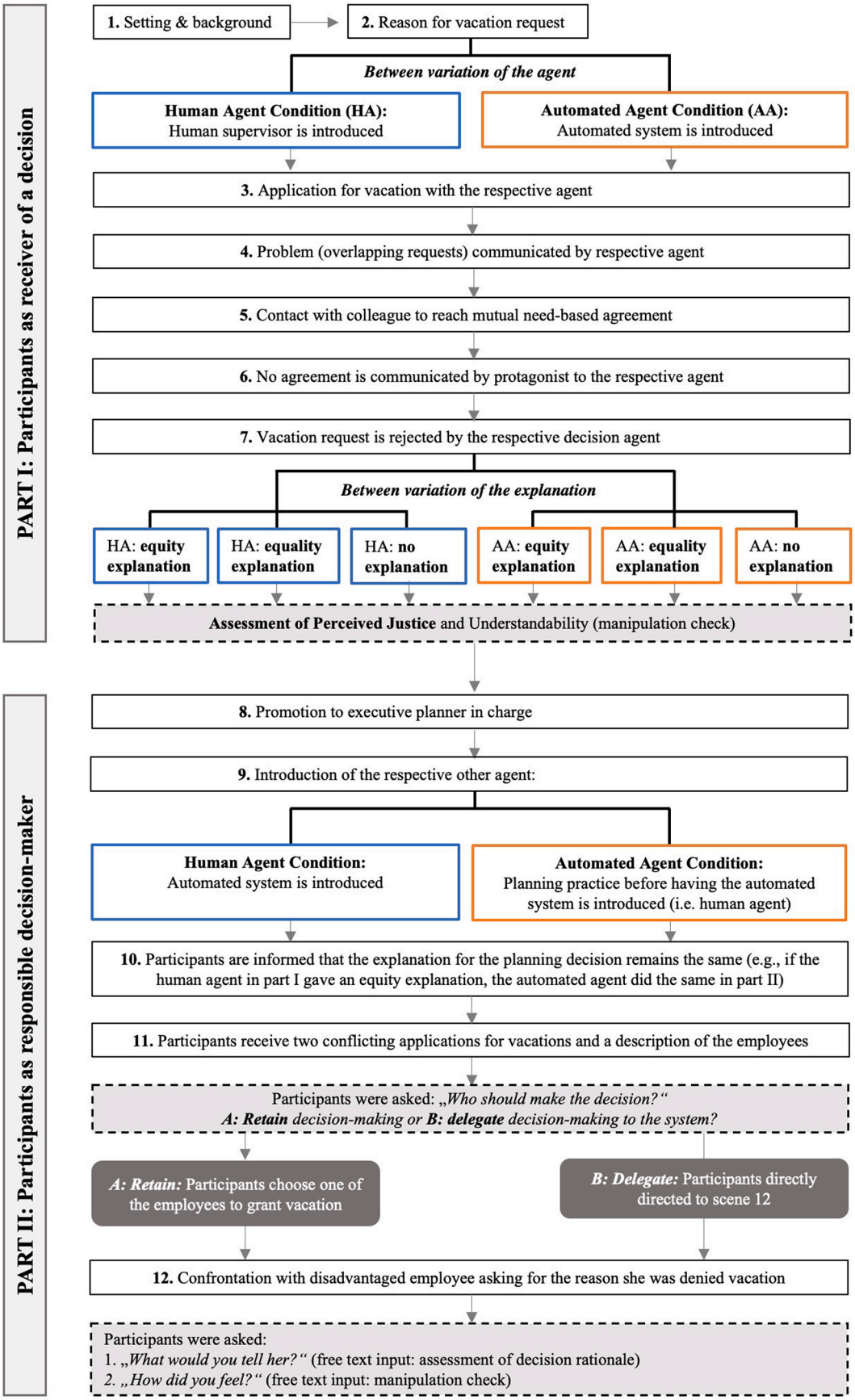


Fig. 1. Overview of the procedure of the experiment and the points of variation between the groups. Note. Numbers indicate the number of the scene described in the procedure section.

Imagine you are a nurse in a hospital. You have been working in the hospital for five years and get along well with your colleagues and your ward manager. It is April and regular vacation planning already took place at the beginning of the year. At that time, you had to plan at least half of your vacation for the year.

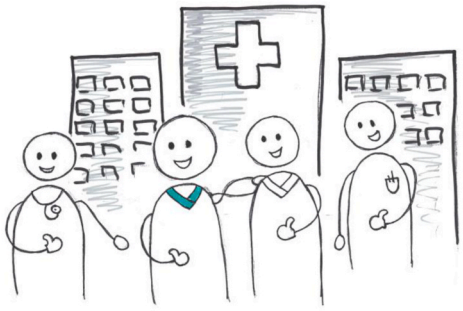


Fig. 2. Example of the comic style storyline presented to participants. Note. The description was translated from German.

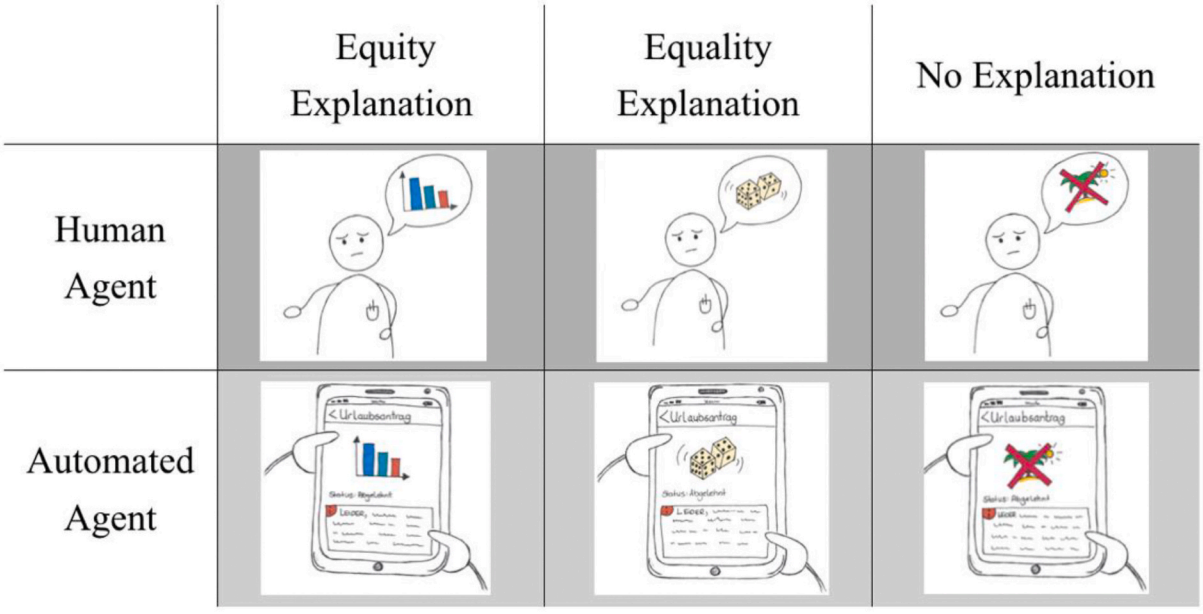


Fig. 3. Graphical representations of the explanation manipulation that participants received in the respective experimental conditions.

used before the automated system was introduced (i.e., that a human decision-maker made decisions) (Scenes 9 and 10). The explanation between the both agents was held consistent throughout the study. This means that the agent introduced at this point used the same explanation as the agent in part one (e.g., participants whose vacation request was rejected by a human-agent and received an equity explanation in part one, were introduced to an automated agent that bases its decisions about conflicting vacation requests on the number of hours employees worked overtime in part two). Accordingly, each participant was aware of the two agents but only one type of explanation. Afterwards, participants were told that two female employees had applied for vacation for the same period of time, but that only one request can be granted (Scene 11). Participants received background information for both employees (e.g., how many children, age). In two pre-studies with first $N = 90$ and second $N = 51$ participants,[5] we tested the background information for the employees. In pre-study 1, we found that participants react strongly to slight differences in background information (e.g., participants predominantly granted vacation to an employee with young children compared to an employee who has to take care of elderly parents). In pre-study 2, we used these insights to ensure that deciding between the two employees would be difficult. Both employee descriptions included similar background information that did not clearly indicate that one of the employees might deserve or need the vacation more than the other. After reading the employees' background information, participants

[5] Results of the pre-studies are available in Appendix C of the supplementary material.

could either decide between the two employees or delegate the decision-making to the automated system.

In Scene 12, participants were confronted by the employee whose vacation request was denied. In an open-ended question, participants were then asked to provide a rationale for their decision to the respective employee and to describe their feelings that were evoked by the employee's disappointment. After the assessment of demographic and control variables, participants were thanked for their participation and encouraged to forward the survey link to colleagues.

3.3. Measures[6]

Justice perceptions. To measure justice perceptions, we used the German version of the Organizational Justice scale (Colquitt et al., 2001; Maier, Streicher, Jonas, & Woschée, 2007) with its four facets: procedural justice (e.g., "Have those procedures of vacation allocation upheld ethical and moral standards?"), distributive justice (e.g., "Does the decision reflect the effort you have put into your work?"), interpersonal justice (e.g., "Have you felt to be treated in a polite manner?"), and informational justice (e.g., "Have the procedures been explained thoroughly?"). Participants responded to the items on a scale from 1 ("not at all") to 5 ("completely"). The wording of the seven procedural justice items was adapted slightly to fit the context of the present study (original: "To what extent have you been able to appeal the (outcome) arrived at by those procedures?", adapted: "To what extent will you be able to appeal the allocation of vacation arrived at by those procedures?"). One of the distributive justice items ("Does the decision reflect what you have contributed to the organization?") was eliminated because it did not fit within the context of the present study resulting in three items that assessed distributive justice. The four items assessing interpersonal justice and the five items assessing informational justice were changed to the passive voice to increase consistency and comparability between the system and the human condition (original: "Has (he/she) been candid in (his/her) communications with you?", adapted: "Have the communications with you been candid?").

Choice to delegate. In part two of the study, participants were asked to respond to the item: "Who should decide?" Participants could choose between "I make the decision by myself" vs. "The system should decide." In this way, we measured participants' choice to delegate.

Decision rationale. Participants responded to the open-ended question "What would you tell Marion/Manuela [the name of the employee whose vacation request was denied]?" These responses were analyzed using the qualitative content analysis method suggested by Mayring (2010, 2014). This method includes eight steps to qualitatively analyze open ended responses: 1. Definition of material; 2. Defining the aim of the analysis; 3. Defining selection criteria; 4. Categorization of synonyms; 5. Similarity categorization; 6. Processing of material; 7. Revision of data; and 8. Interpretation of data. All participant statements were listed in one sheet. First, statements were summarized in one category when they were semantically similar. When participants wrote several sentences, the context was taken into consideration to estimate the semantic similarity. Antonymous statements were summarized in opposite categories. Single statements were subsumed in a residual category to increase reliability (Mayring, 2010). In a second step, it was examined whether further statements could be subsumed within the above categories. Accordingly, all statements were processed again and allocated to broader categories. When all the statements were assigned to categories, definitions and examples for the categories were extracted. The resulting categories (presented in detail in the results section) with their definition and two examples were given to one independent research assistant. She was asked to assign each participant statement to the categories and was allowed to assign one statement (often consisting of several sentences) to one or more categories (Mayring, 2010, 2014). Interrater reliability was calculated to compare the authors' categorization with the categorization of the independent research assistant for each category individually using Cohen's Kappa (Fleiss & Cohen, 1973). This means we assessed whether both coders agreed on the binary classification "category absent vs. present".

3.3.1. Manipulation check

To investigate whether explanations helped participants to understand the decision-making of the respective decision agent to a higher degree, we measured participants' perceived understandability of the agent. Understandability was measured with two (out of the original five) items from Madsen and Gregor's (2000) subscale, perceived understandability. These were: "It is easy to follow what the agent does" and "I know what will happen the next time because I understand how the agent behaves". Participants responded to the items on a scale from 1 ("I don't agree at all") to 5 ("I fully agree").

To examine whether confronting the disadvantaged employee evoked negative feelings, we asked participants in an open-ended question how they felt after providing the respective employee with a rationale for the rejected vacation request. The qualitative analysis followed the same process as the analysis of participants' decision rationale.

3.3.2. Demographical and control variables

We assessed demographic information (gender, age, tenure, whether participants had already been responsible for scheduling, and participants' relationship to their current supervisor on a scale from 1 ["very good"] to 5 ["very bad"]). To control for possible influence of participants' affinity for technology, participants responded to the German version of the Affinity for Technology Interaction scale (Franke et al., 2018, 2019). The scale consists of nine items (e.g., "I like testing the functions of a new technical system") and participants responded to the items on a scale from 1 ("I completely disagree") to 6 ("I fully agree").

3.4. Initial analyses, descriptives, manipulation checks, and analysis information

Before we tested our hypotheses[7], we analyzed scale reliabilities (see Table 1). Cronbach's α for the procedural justice scale was 0.59, less than the acceptable value of 0.70 (Cortina, 1993). Therefore, the reliability of the procedural justice scale would not have allowed us to make conclusions for procedural justice as a uniform construct. We therefore refrained to analyze our hypotheses regarding procedural justice but conducted explorative analyses to investigate potentially underlying reasons for the low reliability of the scale.

Results indicated that providing an explanation increased perceived understandability (see Table 2). Regarding negative feelings and discomfort when participants were required to provide a rationale for the unfavorable decision outcome to the decision-recipient in part two of the study, results indicated that 84.9% of participants described negative affect including the subcategories (a) general descriptions of negative affect, (b) anger, (c) guilt/bad conscience/shame (d) sadness/helplessness, (e) frustration/disappointment, (f) stress/pressure.[8] All subcategories of emotional affect reached good agreement (Cohen's κ ranged from 0.68 to 0.96).

To estimate the interrater reliability between the two raters for the

[6] The items of all scales can be found in Appendix B of the supplementary material.

[7] When controlling for age, gender, tenure, relation to the supervisor, affinity for technology interaction, or whether participants had been responsible for scheduling in the past, the results did not change in a way that would have affected interpretation of the hypotheses. Results can be made available upon request.

[8] Results for the subcategories can be found in Appendix D of the supplementary materials.

Table 1
Means, standard deviations and correlations of all assessed variables.

	Scale	*M* (*SD*)	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.
1.	Agent	0.50 (0.50)	–																
2.	Explanation	0.65 (0.48)	.02	–															
3.	Justice (all)	2.40 (0.61)	-.07	.07	–														
4.	DistrJ	1.83 (0.86)	-.08	-.01	.66**	(.81)													
5.	InterpersJ	2.77 (0.97)	-.26**	-.02	.75**	.29**	(.83)												
6.	InformJ	2.60 (1.01)	.18**	.19**	.73**	.21**	.42**	(.85)											
7.	ProcedJ	2.42 (0.63)	-.05	.00	.67**	.45**	.31**	.31**	(.59)										
8.	PJ 1	2.40 (1.35)	-.29**	.05	.31**	.27**	.17*	.03	.51**	–									
9.	PJ 2	1.89 (1.07)	-.12	-.07	.38**	.38**	.16*	.06	.61**	.41**	–								
10.	PJ 3	2.61 (1.20)	.06	-.09	.44**	.17*	.26**	.32**	.58**	.03	.09	–							
11.	PJ 4	2.84 (1.26)	.29**	-.02	.41**	.11	.25**	.34**	.52**	-.07	.15*	.43**	–						
12.	PJ 5	2.82 (1.08)	-.08	.11	.35**	.18**	.20**	.16*	.56**	.10	.21**	.28**	.26**	–					
13.	PJ 6	2.06 (1.13)	.09	-.01	.12	.15*	-.12	-.01	.45**	.19**	.24**	.06	.08	.10	–				
14.	PJ 7	2.29 (1.09)	-.12	.05	.52**	.45**	.26**	.28**	.56**	.19**	.27**	.29**	.11	.25**	.07	–			
15.	Understand	2.75 (1.12)	.01	.21**	.43**	.16*	.34**	.40**	.29**	.08	.12	.31**	.29**	.26**	-.11	.14*	(.70)		
16.	DelChoice	1.56 (0.50)	-.20**	.12	.19**	.04	.22**	.19**	.08	.13	-.01	-.02	.02	.12	-.08	.12	.10	–	
17.	Age	35.25 (11.86)	.02	-.08	.00	-.07	.00	.04	.04	.05	-.04	.12	.02	-.01	.14	-.14*	-.05	-.10	–

Note. The diagonal shows Cronbach's Alpha reliabilities in italics. Justice (all) = Mean of procedural, distributive, interpersonal, and informational justice. ProcedJ = procedural justice; DistrJ = distributive justice; InterpersJ = interpersonal justice; InformJ = informational justice; PJX = indicates the single items of the procedural justice scale (see Table 2 for the complete item description); DelChoice = Choice to delegate; Coding of agent 0 = human, 1 = automated; Coding of explanation: 0 = no explanation, 1 = explanation (equity and equality combined). Coding of DelChoice: 1 = retain decision-making, 2 = delegate decision-making. $N = 209$.
**$p < .01$, *$p < .05$.

qualitative analyses of participants' decision rationale, Cohen's Kappa (κ) was used. Cohen's κ values less or equal to 0.10 demonstrate no agreement, κ values over 0.10 and up to .40 weak agreement, κ values over 0.40 and up to .60 clear agreement, κ values over 0.60 and up to .80 good agreement, and κ values higher than 0.80 (nearly) complete agreement (Fleiss & Cohen, 1973).

Partial eta squared (η_p^2) were calculated to estimate effect sizes in univariate and multivariate analyses. Partial eta squared values around 0.01 reflect a small, values of 0.06 a medium, and values above 0.14 a large effect size (cf. Richardson, 2011). Cohen's d was calculated to estimate effect sizes in planned contrasts. *d*-values around 0.2 reflect a small, values around 0.5 a medium and values around 0.8 a large effect size (Cohen, 1988). Regarding effect sizes of χ^2-tests, we calculated Cramér's V. Values of Cramér's V are classified as small at around 0.10, as medium above 0.30 and as large above 0.50 (Cohen, 1988; Ellis, 2010).

4. Results

4.1. Hypotheses 1, 2 and 3 as well as Research question 1

To test our hypotheses and RQ1, a 2 (agent: human vs. automated) x 3 (explanation: equity vs. equality vs. no explanation) MANOVA with interpersonal, distributive and informational justice as dependent variables was conducted and followed by univariate ANOVAs to investigate the effects for the single dependent variables. There was a significant effect of agent with a large effect size, *Wilk's* $\Lambda = 0.82$, $F(3, 201) = 14.71$, $p < .01$, $\eta_p^2 = 0.18$, a significant effect of explanation with a medium effect size, *Wilk's* $\Lambda = 0.93$, $F(6, 402) = 2.59$, $p = .02$, $\eta_p^2 = 0.04$, and a significant interaction effect of agent and explanation with a medium effect size, *Wilk's* $\Lambda = 0.92$, $F(6, 402) = 2.81$, $p = .01$, $\eta_p^2 = 0.04$. We then investigated the univariate ANOVA results (means, standard deviations, inferential statistics, and effect sizes are displayed in Table 2). In line with RQ1 and in order to ensure the correct interpretation of the main effects, we always considered interaction effects in the interpretation. Following, we only report interaction results with significant effects. Since we have postulated specific hypotheses, we performed planned contrasts to analyze the differences between single groups of the explanation condition.

Hypothesis 1a stated that participants perceive stronger interpersonal justice in the human compared to the automated agent condition. Results revealed a medium effect, showing that interpersonal justice was significantly stronger in the human ($M = 3.03$, $SD = 0.97$) than in the automated agent condition ($M = 2.52$, $SD = 0.91$), $F(1, 203) = 15.57$, $p < .01$, $\eta_p^2 = 0.07$. This supports Hypothesis 1a.

Hypotheses 1c stated that participants perceive stronger distributive justice in the automated compared to the human agent condition. There was no significant difference between human ($M = 1.89$, $SD = 0.84$) and automated agent ($M = 1.76$, $SD = 0.87$) regarding perceptions of distributive justice, $F(1, 203) = 1.25$, $p = .26$, $\eta_p^2 = 0.01$. Thus, Hypothesis 1c was not supported.

Hypothesis 1d postulated that participants perceive stronger informational justice in the automated compared to the human agent condition. Results revealed a small to medium effect showing that informational justice was significantly stronger in the automated ($M = 2.78$, $SD = 1.02$) than in the human agent condition ($M = 2.41$, $SD = 0.98$), $F(1, 203) = 6.67$, $p = .01$, $\eta_p^2 = 0.03$. However, results for the interaction effect revealed that informational justice was moderately stronger only in the automated compared to the human agent condition when no explanation was provided, $F(2, 203) = 4.97$, $p = .01$, $\eta_p^2 = 0.05$ (see more details regarding this interaction effect results during the analyses for Hypothesis 2 and in Fig. 4). Therefore, Hypothesis 1d was only partly supported. Additionally, the response to RQ 1 is: there was an interaction between the decision agent and the explanation for informational justice.

Hypothesis 2 stated that participants perceive stronger justice when

Table 2
Means and standard deviations for each group and single ANOVA results (including partial η^2).

Variable	Condition						ANOVA		
	HA-EQUITY	HA-EQUAL	HA-NE	AA-EQUITY	AA-EQUAL	AA-NE	AGENT	NORM	AGENT x NORM
	M (SD)	*M* (SD)	*M* (SD)	*M* (SD)	*M* (SD)	*M* (SD)	*F*(1, 203) (η_p^2)	*F*(2, 203) (η_p^2)	*F*(2, 203) (η_p^2)
Understandability	2.99 (1.15)	2.94 (0.94)	2.34 (0.99)	3.06 (1.14)	2.70 (1.13)	2.51 (1.24)	0.00 (0.00)	5.36* (0.05)	0.64 (0.01)
Interpersonal Justice	2.97 (0.99)	3.13 (1.05)	2.99 (0.88)	2.61 (0.95)	2.33 (0.77)	2.61 (0.98)	15.57** (0.07)	0.11 (0.00)	1.22 (0.01)
Informational Justice	2.58 (0.89)	2.85 (1.08)	1.88 (0.72)	2.72 (0.88)	2.82 (1.22)	2.81 (0.96)	6.67* (0.03)	4.59* (0.04)	4.97* (0.05)
Distributive Justice	1.64 (0.79)	2.10 (0.92)	1.95 (0.78)	1.84 (0.73)	1.72 (0.89)	1.73 (0.99)	1.25 (0.01)	0.68 (0.01)	2.18 (0.02)
Procedural Justice (PJ) 1	2.68 (1.25)	2.88 (1.18)	2.82 (1.43)	1.97 (1.28)	2.33 (1.55)	1.75 (1.02)	18.37** (0.08)	1.22 (0.01)	0.75 (0.01)
PJ 2	1.97 (1.22)	1.94 (1.11)	2.16 (1.15)	1.72 (0.88)	1.73 (1.04)	1.83 (1.03)	3.07 (0.01)	0.51 (0.00)	0.05 (0.00)
PJ 3	2.41 (1.10)	2.66 (1.04)	2.55 (1.29)	2.56 (1.32)	2.48 (1.20)	2.97 (1.21)	0.62 (0.00)	1.02 (0.01)	1.05 (0.01)
PJ 4	2.41 (1.23)	2.41 (1.07)	2.58 (1.00)	3.06 (1.29)	3.36 (1.32)	3.19 (1.35)	19.21** (0.09)	0.36 (0.00)	0.41 (0.00)
PJ 5	3.06 (1.15)	2.91 (0.89)	2.76 (1.05)	3.28 (1.03)	2.33 (1.14)	2.56 (0.97)	1.68 (0.01)	5.97* (0.06)	2.44 (0.02)
PJ 6	1.85 (1.02)	1.78 (0.91)	2.18 (1.31)	2.19 (1.28)	2.36 (1.11)	1.94 (1.01)	2.14 (0.01)	0.04 (0.00)	2.49 (0.02)
PJ 7	2.44 (1.26)	2.53 (1.02)	2.32 (1.02)	2.19 (1.06)	2.18 (1.13)	2.11 (1.04)	3.13 (0.02)	0.33 (0.00)	0.08 (0.00)

Note. HA = human agent group, AA = automated agent group, EQUITY = equity explanation, EQUAL = equality explanation, NE = no explanation group, PJ 1 = "Have you been able to express your views and feelings during the procedures of vacation allocation?", PJ 2 = "Have you had influence over the decision arrived at by the procedures of vacation allocation?", PJ 3 = "Have those procedures of vacation allocation been applied consistently?", PJ 4 = "Have those procedures of vacation allocation been free of bias?", PJ 5 = "Have those procedures of vacation allocation been based on accurate information?", PJ 6 = "To what extent will you be able to appeal the allocation of vacation arrived at by those procedures?", PJ 7 = "Have those procedures of vacation allocation upheld ethical and moral standards?". $n_{HA\text{-}EQUITY} = 34$, $n_{HA\text{-}EQUAL} = 32$, $n_{HA\text{-}NE} = 38$, $n_{AA\text{-}EQUITY} = 36$, $n_{AA\text{-}EQUAL} = 33$, $n_{AA\text{-}NE} = 36$. *$p < .05$, **$p < .01$.

an explanation is present than when it is absent. Multivariate contrasts with explanation as independent variable and interpersonal, distributive, and informational justice as dependent variables showed that the no explanation group differed significantly from the groups that received an explanation with a medium effect size, *Wilk's* $\Lambda = .95$, $F(3, 205) = 3.65$, $p = .01$, $\eta_p^2 = 0.05$. Follow-up univariate contrasts, where we corrected *p*-values following the Bonferroni-Holmes method (Holm, 1979), revealed no significant effects for interpersonal (no explanation $M = 2.80$, $SD = 0.94$; explanation $M = 2.76$, $SD = 0.98$), $t(206) = -0.35$, $p = .72$, $d = -0.04$, or distributive justice (no explanation $M = 1.84$, $SD = 0.89$; explanation $M = 1.82$, $SD = 0.84$), $t(206) = -0.14$, $p = .89$, $d = -0.02$. Only for informational justice (no explanation $M = 2.34$ $SD = 0.96$; explanation $M = 2.74$ $SD = 1.02$) we found a significant small to medium effect of receiving an explanation, $t(206) = -0.35$, $p < .01$, $d = 0.40$. In line with Hypothesis 2, perceptions of informational justice towards a human agent were lower when an explanation was absent ($M = 1.88$, $SD = 0.72$) than when an explanation (equity or equality) was present ($M = 2.71$, $SD = 0.99$). However, not in line with Hypothesis 2, perceptions of informational justice towards an automated agent did not differ when an explanation was absent ($M = 2.81$, $SD = 0.96$) or present ($M = 2.77$, $SD = 1.05$). Concluding, Hypothesis 2 was only supported for informational justice for a human decision-agent whereas giving an explanation did not affect the perceptions of informational justice towards an automated agent.

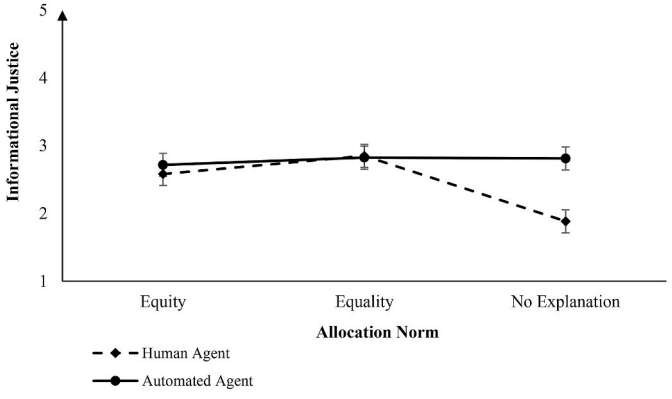


Fig. 4. Interaction between agent and the explanation conditions for informational justice. Note. Error bars indicate standard errors.

Hypothesis 3 stated that participants perceive stronger distributive justice when they receive an explanation referring to the allocation norm of equality than when it refers to equity. Descriptively, the mean values showed differences in the postulated direction with distributive justice being stronger in the equality condition ($M = 1.91$, $SD = 0.92$) than in the equity condition ($M = 1.74$, $SD = 0.76$), but we found no significant difference in a contrast testing this hypothesis, $t(206) = 1.13$, $p = .131$, $d = 0.20$. Hypothesis 3 was therefore not supported.

4.2. Explorative analyses regarding procedural justice

Hypotheses relating to procedural justice could not be tested due to the low reliability of the scale. We, however, explored the results for single procedural justice items using a MANOVA where the seven procedural justice items were entered as dependent variables (see Table 2 for the results of the procedural justice items). The effect of the agent on perceptions of procedural justice was significant with a very large effect size, *Wilks'* $\Lambda = 0.80$, $F(7, 201) = 7.12$, $p < .001$, $\eta_p^2 = 0.20$. Since we did not postulate specific hypotheses for the procedural justice items, alpha-error accumulation of the follow-up ANOVAs was addressed with the Bonferroni-Holm method. Two items reached significance. First, "Have you been able to express your views and feelings during the allocation of vacation?", (Bonferroni-Holm corrected α: .05/7 = 0.007). Ratings for this item were higher for the human than for the automated agent. Second, the item "Has the allocation of vacation been free of bias?" reached significance (Bonferroni-Holm corrected α: 0.05/6 = 0.008). Participants perceived less bias for the automated than for the human agent.

4.3. Explorative analyses regarding the perspective of the decision-maker

RQ 2a asked whether individuals put in the position of a decision-maker prefer to have the authority to make decisions themselves or whether they prefer to delegate the decision to a system. Frequencies of the participants' choice to delegate or retain the decision were compared. A slight majority (56.5%) of participants delegated decision-making to the automated system, whereas 43.5% preferred to retain decision-making. To test whether this difference was significant, a binomial test was conducted. This test was not significant, $p = .072$ suggesting no initial preference for retaining or delegating decision-making.

RQ 2b asked whether preferences to retain or delegate decision-

making depend on the agent that denied the participant's request for vacation in part one of the experiment. Results of a χ^2-test revealed a significant association between the type of decision agent that denied participants' request for vacation in the first part of the survey and participants' choice delegate the decision-making to the system in the second part of the experiment $\chi^2(1) = 8.23$, $p < .01$, $V = 0.20$. This reflects a small to medium effect size. Results showed that participants whose request was denied by a human decision-maker in part one were more likely to delegate their decision-making to the system in part two. Participants whose request was rejected by an automated agent in part one, were more likely to make the decision themselves in part two. The response to RQ 2b therefore is that the preference to retain or delegate decision-making depends on the agent that denied the participant's request for vacation in part one of the experiment.

RQ 3a asked how participants in the position of a decision-maker respond to the employee who asked why she was not granted vacation. Table 3 displays the resulting categories with definitions and examples. As participants often gave responses with multiple rationales, statements could be assigned to several categories. Therefore, frequencies within the categories do not add up to sample size and interrater reliability was calculated for each category. The categories "Obligation to make a decision" and the residual category "Others" did not reach sufficient interrater agreement (see Table 3). These categories were left out in the following analyses and interpretation. We found that participants in the position of a decision agent responded to the employee who was not granted vacation leave by externalizing responsibility, weighting of decision criteria, empathizing with the employee, highlighting their obligation for decision-making, offering alternative solutions, and promising to balance decisions over time.

RQ 3b asked whether responses differ between participants who delegated versus retained the decision-making. We conducted χ^2-tests to compare frequencies. To adjust for alpha-error accumulation the Bonferroni-Holm method (Holm, 1979) was applied ($\alpha = 0.05/5 = 0.01$). There was no significant association between the choice to retain versus delegate the decision and responses of empathy towards the employee. There was a significant association between the choice to retain versus delegate the decision-making and externalization of responsibility in a way that participants who delegated their decision to the automated system were more likely to externalize responsibility. More importantly, participants who delegated the decision-making to the automated system externalized their responsibility in over two thirds of the cases. Furthermore, participants who retained the decision-making more likely referred to weighting available criteria as decision rationale than participants who delegated the decision-making. Additionally, participants who retained decision-making offered alternative solution and wanted to balance the decisions over time more likely than participants who delegated the decision-making. The response to RQ 3b therefore is that we found that decision rationale differed between participants who delegated versus retained the decision-making with respect to the externalization of responsibility, the weighting of available criteria, the offering of alternative solutions, and the balancing of decisions over time.

Finally, to assess possible effects of the kind of explanation (equity vs. equality vs. no explanation), χ^2-tests were conducted for the categories where there was a significant association between delegating and retaining a decision. Results showed no significant effect of explanation.

5. Discussion

The aim of this study was to investigate effects of the decision agent (human vs. automated) and types of explanations (equity vs. equality vs. no explanation) on perceptions of justice. We also explored effects on decision-makers when they can delegate decisions to an automated system. From decision-recipients' perspective, the main result of our study is that justice perceptions differ with respect to decision agents. Perceptions of interpersonal justice were stronger when the decision agent was human. The absence of an explanation only decreased perceptions of informational justice for human but not for automated agents. Additional explorative results revealed that regarding procedural justice, automated agents were perceived as more objective whereas human agents were perceived as providing more voice in decision-making. From the perspective of decision-makers, the most interesting result was that rationales for negative decision outcomes differed between participants in the role of a decision agent who delegated versus the ones who retained decision-making: 70 percent of participants who delegated the decision-making externalized their responsibility. Additionally, they were less likely to refer to potential decision criteria, offer alternative solutions, and less likely to consider balancing decisions over time.

Table 3
Categories resulting from the qualitative analysis of participants' responses to the employee who asked why she was not granted vacation.

Categories of Decision Rationales: definition (examples)	Frequency (%) among delegators ($n = 115$)	Frequency (%) among retainers ($n = 90$)	Frequency (%) overall ($N = 205$)	$\chi^2(1)$	Cohen's κ
A: Externalization of Responsibility: People state that they cannot make the decision themselves or that someone else is responsible. (e.g., "The system has decided so." or "The decision was not mine to make.")	79 (68.7%)	0 (0%)	79 (38.5%)	100.59 ** V = .70	.90
B: Weighting of Criteria: Decision was made based on personal weighting of the criteria (e.g., "Marion gets time off because she has more overtime than you." or "It was decided by chance.")	5 (4.3%)	25 (27.7%)	30 (14.6%)	22.19 ** V = .33	.46
C: Empathizing with Employees: People express understanding, compassion for the feelings of the employee (e.g., "I'm sorry." or "I can understand that you are disappointed.")	61 (53.04%)	36 (40.0%)	97 (47.3%)	3.45 V = .13	.43
D: Obligation to Make a Decision: People state that a decision had to be made and do not explain it further. (e.g., "There is unfortunately no other way." or "I had to make a decision.")	3 (2.6%)	22 (24.4%)	25 (12.2%)	–	.10
E: Offer Alternative Solution: People offer to look for an alternative solution or already provide one (e.g., "We will look for a solution together so that both of you can go on vacation." or "I will take your shift so you can go on vacation.")	21 (18.3%)	28 (31.1%)	49 (23.9%)	4.58 * V = .15	.68
F: Balancing Decisions Over Time: People explain that the respective employee will be preferred in the next decision (e.g., "Next time it will be your turn." or "I will consider you next time.")	15 (13.0%)	29 (32.2%)	44 (21.5%)	11.02 ** V = .23	.96
G: Others: Declarations not covered by previous categories	2 (1.7%)	1 (1.1%)	3 (1.4%)	–	.12

Note. Delegators = Participants who delegated decision-making; Retainers = Participants who retained the decision-making. *$p < .05$, **$p < .01$

5.1. Theoretical implications

5.1.1. Effects of explanations differ for human and automated decision agents

Our investigation of the four facets of justice (Colquitt, 2001; Colquitt et al., 2001) revealed that a holistic view on justice is important when comparing human and automated decision agents. Regarding informational justice, in line with previous research, explanations strengthened informational justice perceptions in the case of a human decision agent (Greenberg & Colquitt, 2013; Skarlicki et al., 2008). However, the absence of an explanation did not affect perceptions of informational justice when the decision was provided by an automated agent. These results contradict research in human-computer interaction either finding or claiming that explanations should lead to stronger justice perceptions (Dodge et al., 2019; Doshi-Velez et al., 2017; M. K. Lee et al., 2019). Notably, explanations led to more understanding regarding the underlying decision-making process in both the human and automated agent condition. However, this also indicates that the difference in perceptions of informational justice between human and automated agents in the absence of an explanation did not depend on the understandability of the decision-making process.

Consequently, the relationship between explanations and justice seem to differ for human and automated agents. We see at least two possible interpretations for this. First, people' s expectations regarding explanations may differ between automated and human agents. Individuals might only expect explanations from human agents (Greenberg & Colquitt, 2013; Skarlicki et al., 2008). This may be due to the fact that people attribute the absence of an explanation differently for human compared to automated agents. In terms of attribution theory, this means that the absence of an explanation in human agents may be attributed internally while for automated agents it may be attributed externally (Kelley & Michela, 1980). For instance, if a human agent does not provide an explanation, this might be attributed to motivational, intentional or situational causes, for instance, not wanting to give an explanation or strategically concealing information (Ananny & Crawford, 2018; Schnackenberg & Tomlinson, 2016). If an automated agent does not provide an explanation, individuals may assume that technical constraints do not allow to provide an explanation. In any case, people might assume that motivation, intention, and situation do play less of a role for automated agents but that its design and related capabilities determine its outputs (Elsbach & Stigliani, 2019). Second, individuals might assume that system decisions rely on complex calculations or algorithms which they would not understand even if they had been provided (Elsbach & Stigliani, 2019; M. K. Lee et al., 2019). These tentative interpretations call for future research to investigate whether, how, and under what conditions explanations and information regarding automated agents' decision-making processes and outputs will affect justice perceptions.

5.1.2. Automated agents influence perceptions of procedural and interpersonal justice

Our findings regarding procedural justice provided several important insights. Specifically, the measure of procedural justice suffered from low internal consistency. This is remarkable, because the procedural justice scale (Colquitt, 2001) is frequently used and validated. Furthermore, Ötting and Maier (2018) also used the scale of Colquitt (2001) when comparing human and automated agents and did not find low internal consistency. Investigating the items more closely revealed that there were two strong differences between human and automated agents. Participants perceived that humans allow for more voice whereas they perceived automated agents as less biased (this pattern also explains the scale' s low internal consistency). One interpretation would be that participants perceived humans to be more adaptable and to offer more potential for control by decision-recipients in decision situations (Kellogg, Valentine, & Christin, 2020). At the same time, automated decision-agents were perceived as providing more objective decision-making processes, which are less prone to biases. These results contribute empirical evidence to previous research which had posited that people ascribe certain characteristics to automated systems (Elsbach & Stigliani, 2019). Our study encourages future research to investigate presumable automatic assumptions or heuristics regarding characteristics of automated systems.

Furthermore, the effects between decision agents regarding procedural justice contradict previous research findings that showed no differences between human and automated agents in scheduling contexts (M. K. Lee, 2018; Ötting & Maier, 2018). We found different results potentially because participants in our study were people with actual domain knowledge and because we used an immersive storytelling procedure. We thus tentatively conclude that the current findings more accurately reflect people' s reactions to automated systems in applied settings.

Overall, results for procedural justice point towards three insights. First, when comparing justice between humans and automated agents, it might be necessary to develop new ways to assess and compare justice with respect to automated and human decision agents. This could be comparable to research on trust in automation, where the main concepts and processes regarding trust in humans and trust in automated systems seem comparable (e.g., trustworthiness, trust, reliance) whereas ways to measure or define specific concepts (e.g., facets of trustworthiness) might need to be adapted to adequately measure people' s perceptions of automated systems (Madhavan & Wiegmann, 2007). In the case of justice, main concepts and processes (e.g., perceptions of justice consist of several facets; justice positively affects organizational outcomes; Ötting & Maier, 2018) also seem comparable, but ways to measure and define specific concepts might need adaptation when they are applied in the context of automated decision-making (e.g., what does procedural justice mean in relation to automated decisions?). Second, there might be stronger effects of decision agents on justice in real world contexts, in which more emotions and stronger uncertainty is involved (Colquitt et al., 2012; Cremer & Sedikides, 2005; De Cremer & Van Hiel, 2008; Van den Bos, 2001). Third, people inherently perceive humans as more flexible and offering more voice (Folger, 1977; Lind et al., 1990; Tangirala & Ramanujam, 2008), whereas they perceive automated agents as more consistent and less biased (Mouloua, 2018; Skitka et al., 2000).

Our interpretation that automated agents were perceived as more consistent but leaving less room for complaints and interpersonal communication are also supported by our results regarding interpersonal justice. In line with previous research, individuals feel better understood and respected by a human than by an automated agent (M. K. Lee, 2018; Patient & Skarlicki, 2010). Furthermore, results for interpersonal justice might also indicate that people seem to hope for more individualization in human decision-making processes, whereas automated decision-making processes are perceived as consistent and invariable (see also Longoni et al., 2019). Ultimately, having voice, less consistency, and more interpersonal closeness might also lead to higher perceptions of control over decision processes with human decision-makers involved Kellogg et al., 2019. This might be due to people assuming more persuasive powers over other people or due to the hope for possible exceptions favoring oneself when humans are responsible for decision-making (Dietvorst & Bharti, 2020). However, this interpretation clearly calls for further examination.

Regarding distributive justice, the results did not support our Hypothesis. Distributive justice was not affected by the different agents and there were no effects of the types of explanations. This might have been due to the specific items that we used to assess distributive justice (Colquitt, 2001). The items mainly reflected the personal contribution at the workplace and its impact on the decision reached. Therefore, further research should assess distributive justice in applied settings where employees have contributed to the organization's success.

5.1.3. Delegating decisions to systems affects decision rationales

Delegating decisions to automated systems or retaining the decision-

autonomy led to a variety of insights regarding participants' rationales for denying one of the employee's vacation request. Specifically, participants tended to delegate their decision less likely to an automated system when they experienced rejection by an automated decision agent in part one of the story. Furthermore, when participants delegated decisions to the system, they were less likely to provide reasons for why the respective employee's vacation request was rejected, less likely to consider possible alternatives or compromises, and less likely to consider ways to offer future compensation for the negative outcome. This might imply that those participants who delegated the decision engaged in less thorough reasoning about the decision, thus being less able to cite specific criteria or thinking about different solutions. Furthermore, nearly 70 percent of participants who delegated their decision also externalized responsibility either mentioning that they were not responsible for the decision or referring to the automated system as being responsible for the decision. Apparently, when delegating their decision, people were also prone to "delegate" responsibility. One interpretation for this finding could be that people believe they can partly make systems accountable for decisions (Bigman et al., 2019). Research has shown that people tend to not question automated decisions because they assume that system decision-making is evaluated and checked by system designers or responsible parties in the organizations (J. D. Lee & See, 2004; Parasuraman & Riley, 1997). People might assume that the use of systems and thus also system outputs are approved by a superior instance, and might less likely question the decision outcome (Lammers et al., 2012). This finding is also in line with work characteristics research proposing that people perceive less responsibility for their work outcomes when they perceive less decision autonomy (Hackman & Oldham, 1976; Morgeson et al., 2012) or control Kellogg et al., 2019.

However, it is important to highlight that even if people seem to use the automated system as an excuse for an unfavorable decision, in cases where there are actual consequences of decisions (in high-stakes decision contexts, e.g., medical diagnoses (Longoni et al., 2019)) the automated agent itself cannot share responsibility with the human decision-maker. Thus, delegating decision-making to an automated agent might result in an illusion of shared responsibility. This might be a fruitful direction for future research and feeds into the continuing debate on the responsibility and accountability of AI (Dignum, 2017; Floridi et al., 2018). Consequently, when decisions are (partly) automated, it might be necessary to clearly communicate responsibility and accountability with regard to (semi-)automated decisions (Jobin et al., 2019). Further research needs to examine how to explain responsibilities clearly to users, and to investigate whether clear responsibilities increase the motivation of decision-makers to question decisions of automated agents.

5.2. Limitations

At least four limitations should be considered when interpreting our findings. First, results are based on an imagined situation experienced in an online survey potentially undermining the external validity of the results. However, the story-methodology introduced in this study tried to enhance psychological fidelity and might offer a viable tool for research to investigate the effects of automated agents (Briggs et al., 2012; Diefenbach, Chien, Lenz, & Hassenzahl, 2013). A visual demonstration depicts how an automated agent is to be integrated in real working environments of specific user groups. This could help to increase consistency between participants' conceptions in comparison to text-only vignettes. Differing findings compared to previous studies (e.g., Ötting and Maier (2018) found no effects of the decision-agent and procedural justice did not suffer from low internal consistency) could indicate that participants were more immersed within the current study design. Hence, the experimental manipulation in the present study could have been particularly convincing, which might enhance generalizability of the findings. Whether these differences were actually due to the use of comics or due to other conceptual differences could be investigated in future studies.

Second, the interpretation of the current results is limited due to the specific sample consisting of healthcare professionals in Germany and therefore mainly of female participants (as is the case for German healthcare workers). Thus, the results might not generalize to other contexts. However, the study setting is highly relevant for the healthcare context. Moreover, the current study used a commonly investigated setting (i.e., scheduling, M. K. Lee, 2018; Ötting & Maier, 2018) and therefore results might be applicable to scheduling in different domains.

Third, whereas we tried to keep the storylines for the human and automated decision agent as parallel as possible, the experimental conditions might have not only varied in the decision agent but also in the way of how a decision was communicated to decision-recipients. Clearly, a decision made by a human agent could also be conveyed using a system (e.g., via e-mail) and the decision outcome by an automated system might be communicated to decision-recipients by a human. Effects might thus not exclusively be due to the decision agent and could also be grounded in the way the decision was communicated. Future research could thus also investigate whether there are additional effects when varying not only the decision-agent but also the communication channel. For instance, strong negative reactions regarding interpersonal justice (e.g., feeling disrespected) might emerge when a human agent, who is capable of personal communication, delivers bad news via mail or other communication technology (Timmerman & Harrison, 2005).

Fourth, the findings of the interaction between agent and explanation, as well as the findings regarding the perspective of the decision-maker are explorative in nature. In order to ensure the replicability, Hypothesis-based investigations must be carried out. Nevertheless, the current explorative findings might inspire future studies.

5.3. Main practical implications

Given the present findings, practitioners should be aware that different criteria might be used when the justice of automated versus human agents is evaluated. Employees might not perceive a lack of justice when automated agents do not provide explanations accompanying their outputs that are then translated to decisions. Hence, it is possible that people are less prone to question automated decisions. As a consequence, this might lead to a lack of examination of the decision process, thus making the discovery of system errors and biases less likely. This is especially problematic when questions of accountability are not clarified (Bigman et al., 2019; M. K. Lee et al., 2019), when systems are less than perfectly reliable, when outputs reflect biases in underlying training data (Bellamy et al., 2018; Hacker, 2018), and when automated decisions take over ethically sensitive decisions (Wesche & Sonderegger, 2019). Preventive actions may range from information and education, to ethical guidelines and legal regulations (Langer & König, 2021) aimed at increasing awareness of capabilities and limitations of automated systems and AI in practice.

5.4. Conclusion

The automation of decision-making affects decision-recipients and decision-makers alike. Management and healthcare are just two of many fields where investigating effects on justice, perceived responsibility, as well as explanations and processing of further decision-related information is crucial when automated agents based on AI and machine learning support or fully automate decisions. In this line of research, it will be important to investigate whether underlying concepts (e.g., contributing to justice, trust; Madhavan & Wiegmann, 2007) or well-known relations (e.g., between explanations and justice) may differ when systems become decision agents. Specifically, we must verify expectations and assumptions derived from research in the field of interpersonal decision-making and should not assume that they will

generalize for the context of automated and augmented decision-making.

Credit author statement

Nadine Schlicker: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Markus Langer: Conceptualization, Formal analysis, Writing – review & editing, Supervision, Validation, Sonja Ötting: Conceptualization, Writing – review & editing, Validation, Kevin Baum: Validation, theory, review, Cornelius König: Conceptualization, Validation, review, Dieter Wallach: Validation, review, acquisition of participants.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chb.2021.106837.

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